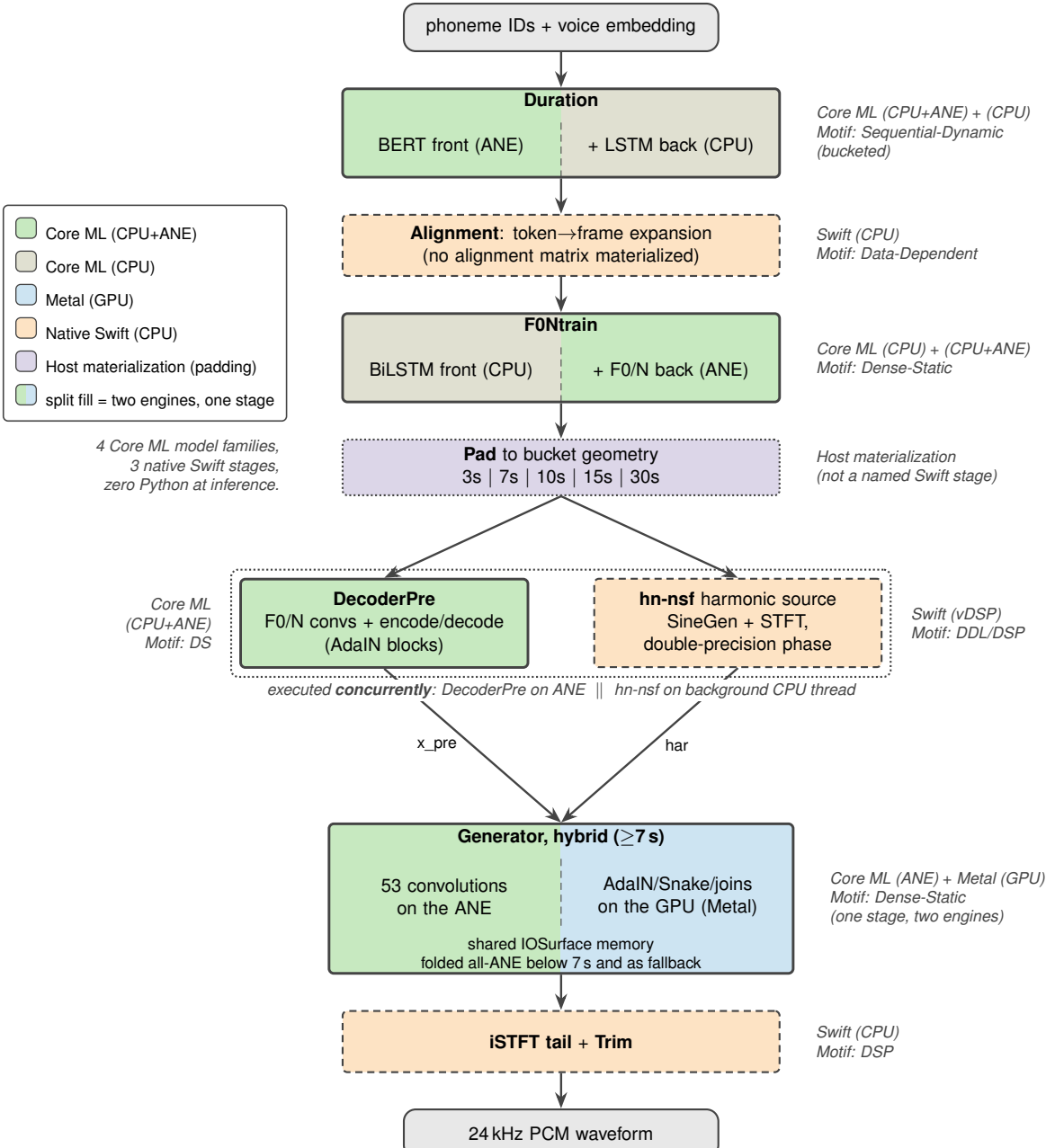
phoneme IDs + voice embedding
Duration
BERT front (ANE)
+ LSTM back (CPU)
Core ML (CPU+ANE) + (CPU)
Motif: Sequential-Dynamic
(bucketed)
Alignment: token→frame expansion
(no alignment matrix materialized)
Swift (CPU)
Motif: Data-Dependent
F0Ntrain
BiLSTM front (CPU)
+ F0/N back (ANE)
Core ML (CPU) + (CPU+ANE)
Motif: Dense-Static
Pad to bucket geometry
3s | 7s | 10s | 15s | 30s
Host materialization
(not a named Swift stage)
Core ML (CPU+ANE)
Core ML (CPU)
Metal (GPU)
Native Swift (CPU)
Host materialization (padding)
split fill = two engines, one stage
4 Core ML model families,
3 native Swift stages,
zero Python at inference.
Core ML
(CPU+ANE)
Motif: DS
DecoderPre
F0/N convs + encode/decode
(AdaIN blocks)
hn-nsf harmonic source
SineGen + STFT,
double-precision phase
Swift (vDSP)
Motif: DDL/DSP
executed concurrently: DecoderPre on ANE || hn-nsf on background CPU thread
x_pre
har
Generator, hybrid (≥7 s)
53 convolutions
on the ANE
AdaIN/Snake/joins
on the GPU (Metal)
shared IOSurface memory
folded all-ANE below 7 s and as fallback
Core ML (ANE) + Metal (GPU)
Motif: Dense-Static
(one stage, two engines)
iSTFT tail + Trim
Swift (CPU)
Motif: DSP
24 kHz PCM waveform